COMPLEX ENGINEERING PROBLEM PROPOSAL

SHOPPING MANAGEMENT SYSTEM

BATCH 2023

SECTION B

GROUP MEMBERS	ROLL NUMBER
MASHOOD HUSSAIN	23K6065
MALIK ABDUL RAHMAN	23K6064
HAMMAD SHAH	23k-6106
FARHAN ALI	23K6034

Course Instructor: Engr. Haris Mohsin

Lab Instructor: Engr. Usama bin Umar

DEPARTMENT OF ELECTRICAL ENGINEERING

NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES – FAST

KARACHI CAMPUS

1. Abstract:

The Shopping Management System (SMS) is an all-inclusive digital solution designed for the efficient administration of product inventories, pricing, and retail transactions. Featuring a dual-interface design, the system provides tailored functionalities for both Administrators (Managers) and End-Users (Customers). The primary objectives are to modernize the consumer shopping experience, streamline executive backend tasks, and promote sustainable digital record-keeping. By integrating automated file handling and structured data management, this system addresses the inherent inefficiencies of traditional manual inventory methods.

2. Introduction:

In today's rapidly evolving technological landscape, businesses must adopt innovative approaches to streamline operations and enhance customer satisfaction. The retail sector, in particular, faces constant challenges in maintaining inventory accuracy and providing seamless checkout experiences. This project introduces a sophisticated Management System designed to bridge the gap between complex backend logistics and user-friendly frontend interactions, providing a scalable solution for businesses of varying sizes.

3. Problem Statement & Sustainable Development Goals (SDG):

The Problem

Traditional retail management often suffers from fragmented data, slow billing processes, and difficulty in real-time inventory tracking. Store owners require a centralized tool to manage detailed product information and generate invoices instantly to reduce customer wait times.

Alignment with SDGs: This project is mapped to SDG 9: Industry, Innovation, and Infrastructure.

Objective: To build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation through digital transformation in the local retail sector.

4. Literature Review:

A comprehensive review of existing retail methodologies reveals a significant shift from manual ledger-based systems to automated Point of Sale (POS) software. Current research emphasizes the importance of data persistence and user-centric design. This project identifies a gap in lightweight, file-based systems that do not require heavy database overhead, making it ideal for small to medium-sized enterprises (SMEs) looking for low-cost digital transitions.

5. Project Goals:

- **Connectivity**: Build a seamless digital bridge between customers and product catalogs.
- **Management**: Empower store owners to organize product details and pricing efficiently.
- **Efficiency**: Reduce the purchasing journey to a few simple steps.
- **Accounting**: Automate the generation and storage of digital invoices/bills.

6. System Architecture & Methodology:

The methodology or architecture employed in developing the Shopping Management System is detailed in this section. Discussion includes the key technologies, frameworks, or algorithms utilized to implement the system, clarifying how the architecture supports scalability and elasticity.

1. Initialization:
    a. The code begins by defining constants and function prototypes.
    b. It declares arrays to store product names, prices, and related information, as well as arrays for the customer's shopping cart.
2. Menu Display:
    a. There's a function `displayMenu` that prints a menu for both managers and customers.
3. File Handling:
    a. The program reads product information from a file ("products.txt") and writes back to it. This enables data persistence across program executions.
4. Manager Interface:
    a. If the user chooses the manager interface, they are prompted for a password ("admin123" in this case).
    b. Once authenticated, the manager can perform actions like adding products, displaying products, removing products, updating prices, and exiting the program.
5. Customer Interface:
    a. If the user chooses the customer interface, they can display products, add customer details, add products to a cart, print a bill, and exit.
6. 6.Functions:
    a. Functions like `addProduct`, `displayProducts`, `removeProduct`, `updatePrices` allow the manager to perform various tasks related to product management.
    b. Functions like `addToCart` and `printBill` are for customer-related tasks.
7. Discounts:
    a. The `applyDiscount` function calculates a discount based on the customer's age.
8. File Output for Bill:
    a. The program writes the bill information to a file ("bill.txt") in an organized manner.
9. Looping:
    a. The program uses loops to keep the user in the interface until they choose to exit.
10. User Input Handling:
    a. Input is received using `cin` and validated at various points in the program.
11. Clearing Input Buffer:
    a. `cin.ignore()` is used to clear the input buffer, ensuring proper input handling.
12. Code Organization:
    a. The code is organized into functions, enhancing modularity and readability.

13. Improvements:
    a. Error handling for incorrect inputs is limited; adding more robust input validation would enhance the program's reliability.
    b. Security considerations, such as handling passwords securely, could be improved.

Remember that for production-level applications, additional considerations for security, input validation, and error handling should be implemented.

7. Project Milestones and Deliverables:

Phase I: Requirement Analysis and Logic Flowcharting.

Phase II: Development of the Core Engine and File Handling systems.

Phase III: UI/UX Design for the Command Line Interface (CLI).

Phase IV: Testing, Debugging, and Security Refinement.

8. Work Division:

A. • **Data entry/input codes**:
Mashood.
Malik Abdul Rahman.
B. **Display data/output codes**:
Hammad Shah.
Farhan Ali

9. Costing:

Zero

10. Conclusion:

This system demonstrates how structured programming streamlines retail operations through modular inventory and billing logic. While robust, future iterations will integrate advanced encryption and a GUI to further enhance security and user experience.

Mashood 2nd. Report

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